

Capturing TCP and TLS Handshake Using Network Namespaces

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Aim:

To create two network namespaces (red and blue), establish a secure HTTPS connection between them, capture the TLS 3-way handshake, and analyze the captured packets in Wireshark.

Tools Used:

Linux Network Namespaces, OpenSSL, curl, tcpdump, Wireshark

Description:

Two network namespaces named **red** and **blue** were created and connected using virtual interfaces. The blue namespace acted as server. The red namespace acted as the client and initiated an HTTPS request using the curl command. During this communication, tcpdump was used in the blue namespace to capture the packets and save it into a **.pcap** file. The captured file was later analyzed using Wireshark.

Commands Used

Certificate Generation:

```
openssl genrsa -out server.key 2048
openssl req -new -x509 -key server.key -out certificate.crt -days 365
```

Starting TLS Server in Blue Namespace:

```
sudo ip netns exec blue openssl s_server -accept 443 \
-cert certificate.crt -key server.key
```

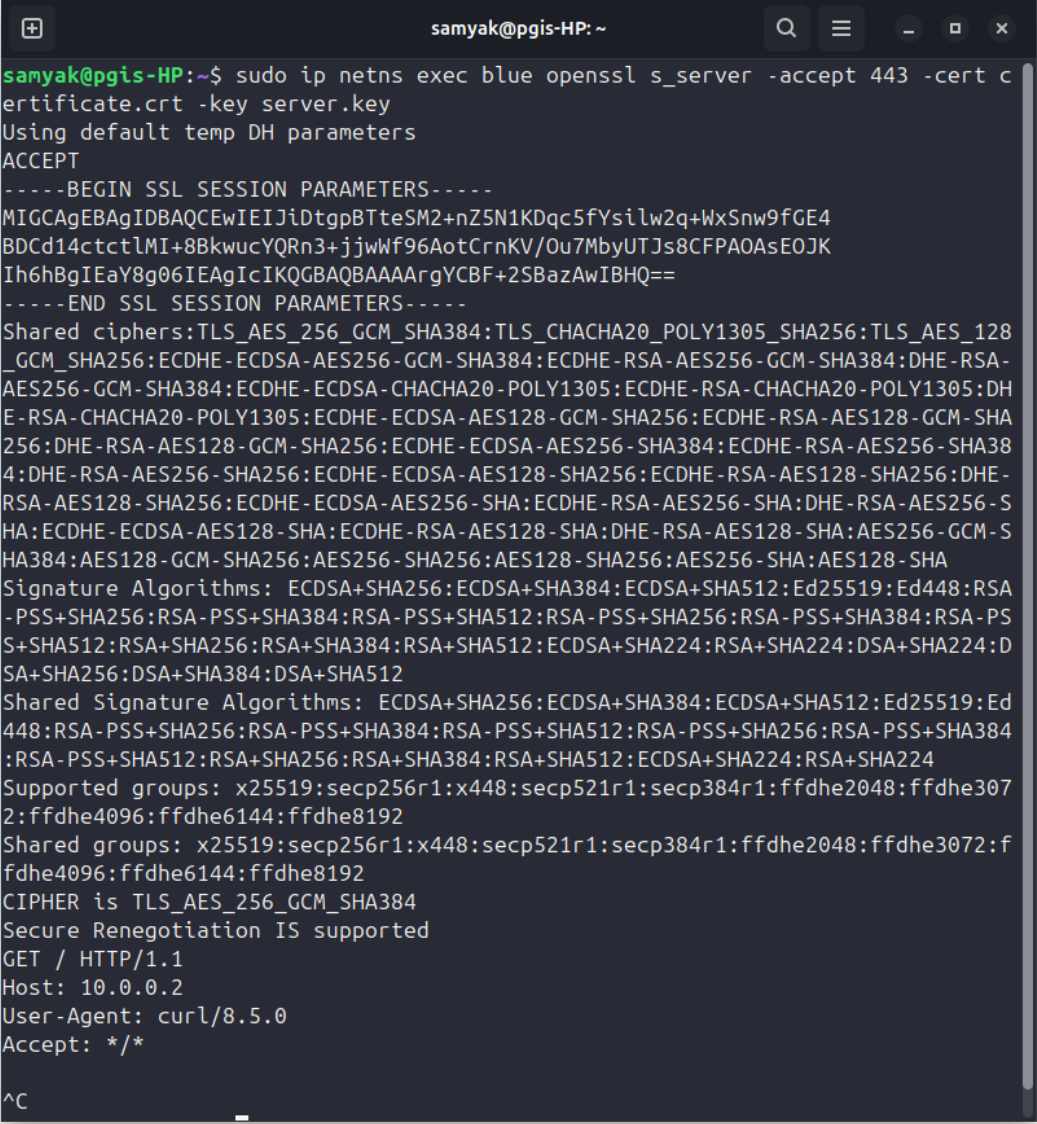
Capturing Packets:

```
sudo ip netns exec blue tcpdump -i eth1 -w tls_capture.pcap
```

Generating HTTPS Request from Red Namespace:

```
sudo ip netns exec red curl -k https://10.0.0.2
```

Screenshots:

A terminal window titled 'samyak@pgis-HP: ~' with search, menu, and window control icons. The terminal shows the execution of 'sudo ip netns exec blue openssl s_server -accept 443 -cert certificate.crt -key server.key'. The output includes 'Using default temp DH parameters', 'ACCEPT', '-----BEGIN SSL SESSION PARAMETERS-----', a long base64-encoded string, '-----END SSL SESSION PARAMETERS-----', a list of shared ciphers, signature algorithms, supported and shared groups, and the selected cipher 'TLS_AES_256_GCM_SHA384'. It also shows 'Secure Renegotiation IS supported' and an incoming 'GET / HTTP/1.1' request from 'Host: 10.0.0.2' using 'User-Agent: curl/8.5.0'. The prompt 'Accept: */*' and a carriage return '^C' are also visible.

```
samyak@pgis-HP:~$ sudo ip netns exec blue openssl s_server -accept 443 -cert c
ertificate.crt -key server.key
Using default temp DH parameters
ACCEPT
-----BEGIN SSL SESSION PARAMETERS-----
MIGCAgEBAgIDBAQCEwIEIjIDtgpBTteSM2+nZ5N1KDqc5fYsIlw2q+WxSnw9fGE4
BDCd14ctctlMI+8BkwucYQRn3+jjwWf96AotCrnKV/Ou7MbyUTJs8CFPA0AsEOJK
Ih6hBgIEaY8g06IEAgIcIKQGBAQBAARgYCBF+2SBazAwIBHQ==
-----END SSL SESSION PARAMETERS-----
Shared ciphers:TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128
_GCM_SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384:DHE-RSA-
AES256-GCM-SHA384:ECDHE-ECDSA-CHACHA20-POLY1305:ECDHE-RSA-CHACHA20-POLY1305:DH
E-RSA-CHACHA20-POLY1305:ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA
256:DHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-SHA384:ECDHE-RSA-AES256-SHA38
4:DHE-RSA-AES256-SHA256:ECDHE-ECDSA-AES128-SHA256:ECDHE-RSA-AES128-SHA256:DHE-
RSA-AES128-SHA256:ECDHE-ECDSA-AES256-SHA:ECDHE-RSA-AES256-SHA:DHE-RSA-AES256-S
HA:ECDHE-ECDSA-AES128-SHA:ECDHE-RSA-AES128-SHA:DHE-RSA-AES128-SHA:AES256-GCM-S
HA384:AES128-GCM-SHA256:AES256-SHA256:AES128-SHA256:AES256-SHA:AES128-SHA
Signature Algorithms: ECDSA+SHA256:ECDSA+SHA384:ECDSA+SHA512:Ed25519:Ed448:RSA
-PSS+SHA256:RSA-PSS+SHA384:RSA-PSS+SHA512:RSA-PSS+SHA256:RSA-PSS+SHA384:RSA-PS
S+SHA512:RSA+SHA256:RSA+SHA384:RSA+SHA512:ECDSA+SHA224:RSA+SHA224:DSA+SHA224:D
SA+SHA256:DSA+SHA384:DSA+SHA512
Shared Signature Algorithms: ECDSA+SHA256:ECDSA+SHA384:ECDSA+SHA512:Ed25519:Ed
448:RSA-PSS+SHA256:RSA-PSS+SHA384:RSA-PSS+SHA512:RSA-PSS+SHA256:RSA-PSS+SHA384
:RSA-PSS+SHA512:RSA+SHA256:RSA+SHA384:RSA+SHA512:ECDSA+SHA224:RSA+SHA224
Supported groups: x25519:secp256r1:x448:secp521r1:secp384r1:ffdhe2048:ffdhe307
2:ffdhe4096:ffdhe6144:ffdhe8192
Shared groups: x25519:secp256r1:x448:secp521r1:secp384r1:ffdhe2048:ffdhe3072:f
fdhe4096:ffdhe6144:ffdhe8192
CIPHER is TLS_AES_256_GCM_SHA384
Secure Renegotiation IS supported
GET / HTTP/1.1
Host: 10.0.0.2
User-Agent: curl/8.5.0
Accept: */*
^C
```

Figure 1: Captured traffic at server and stored in a pcap file

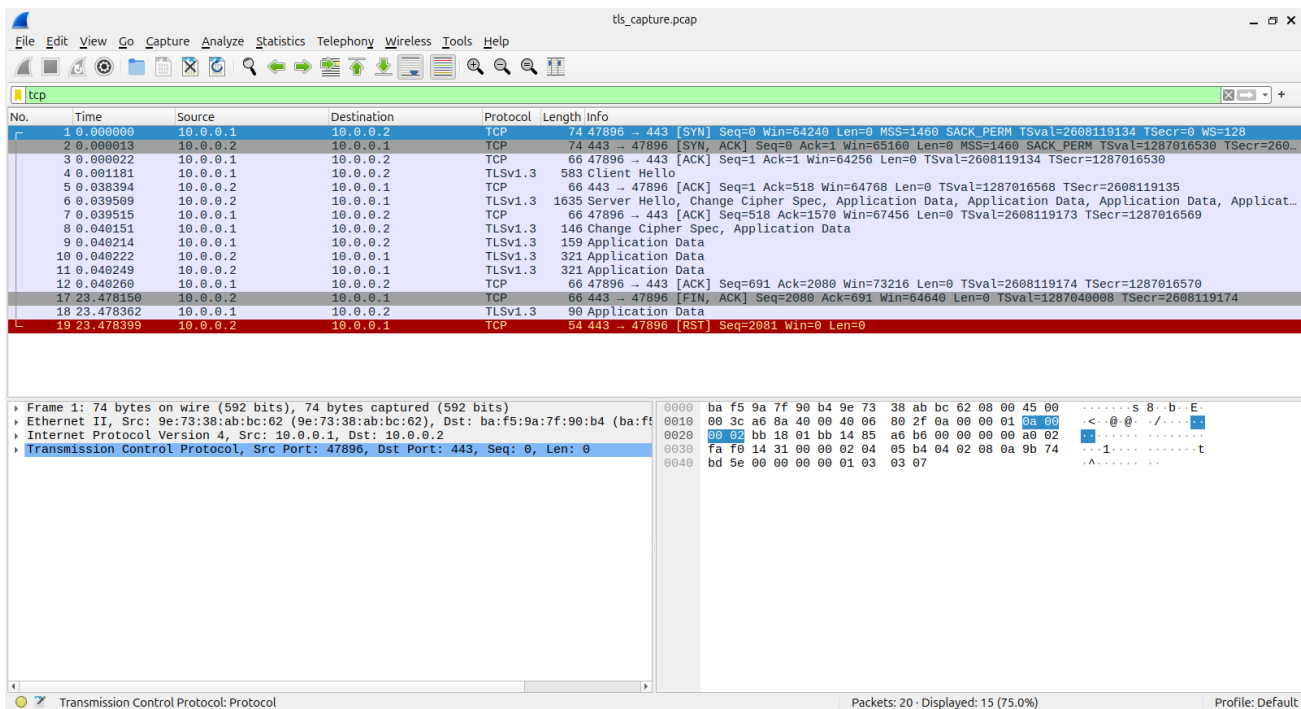


Figure 2: Opening pcap in Wireshark

This is a Wireshark capture of a TLS 1.3 connection between:

- Client: 10.0.0.1
- Server: 10.0.0.2
- Service Port: 443 (HTTPS)

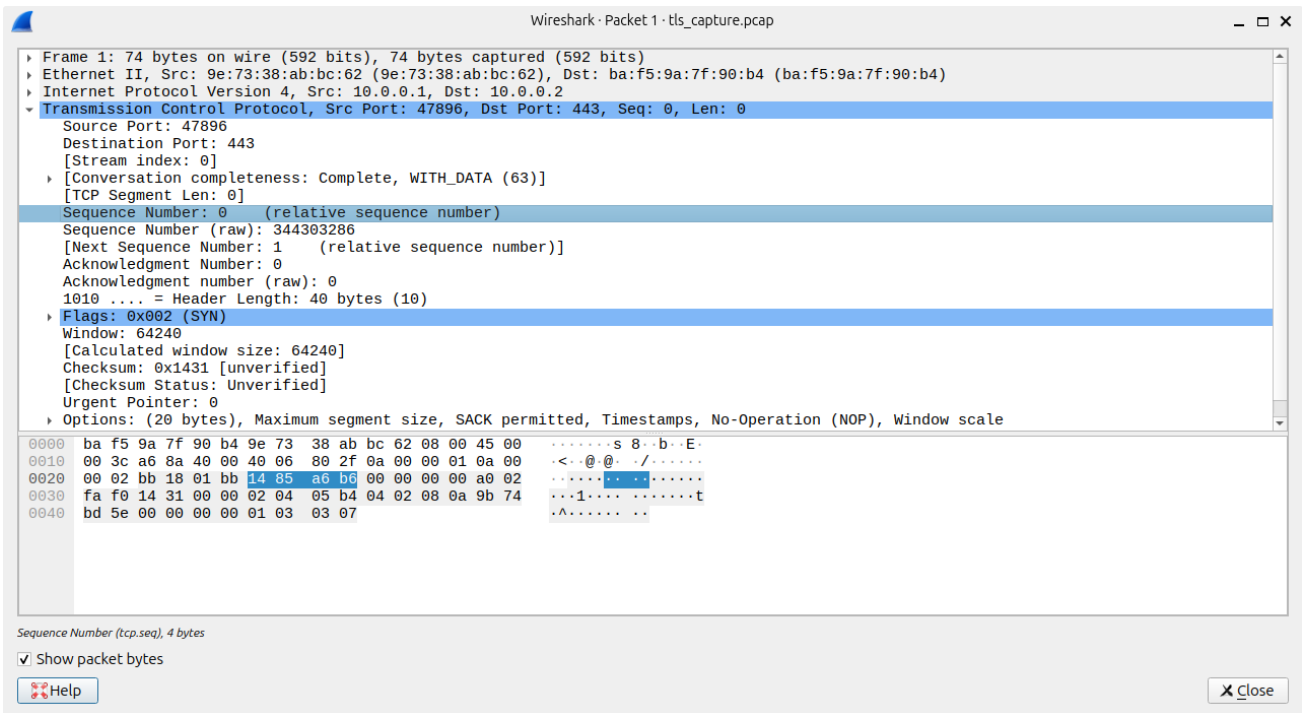


Figure 3: TCP Initial SYN

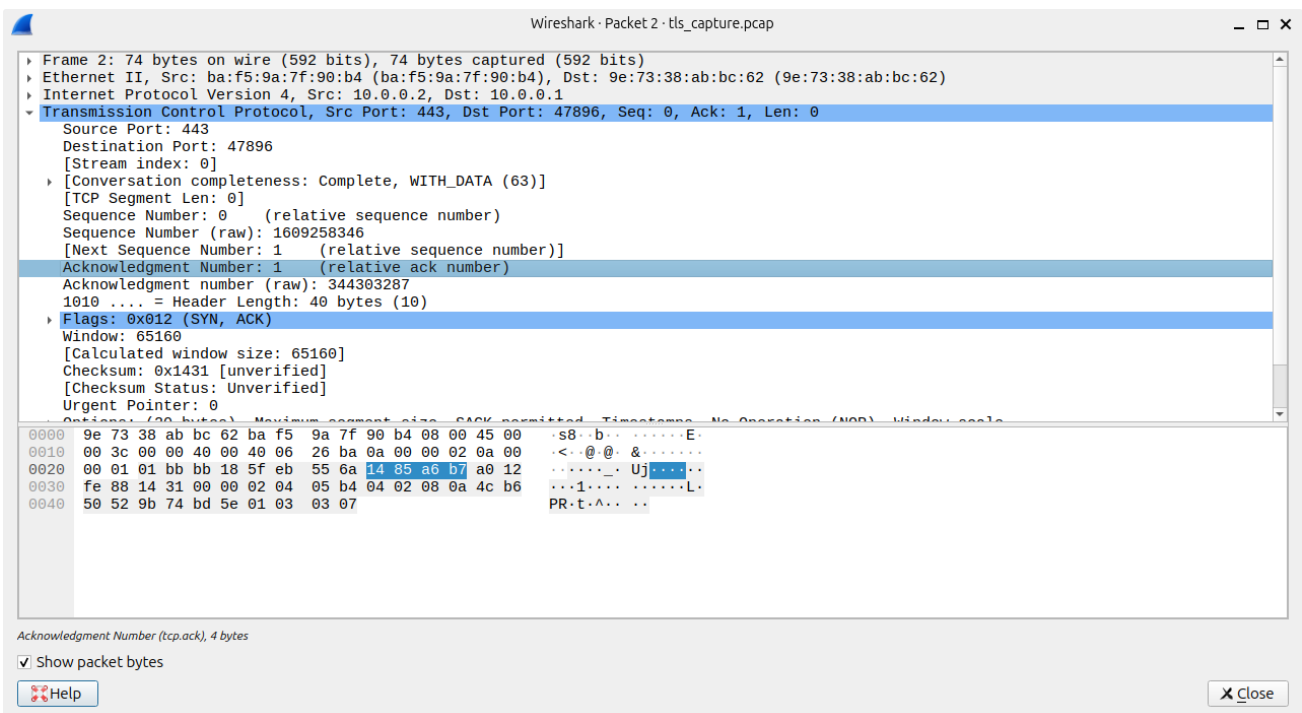


Figure 4: TCP SYN-ACK

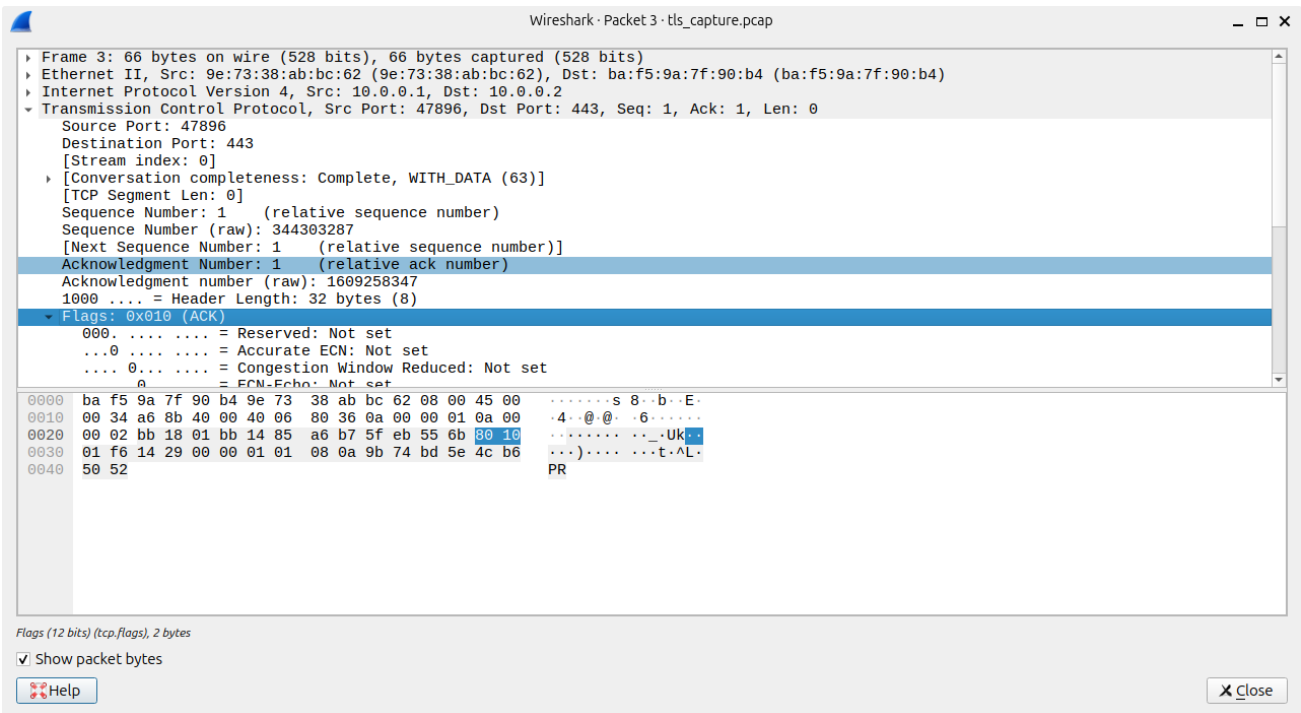


Figure 5: TCP Final ACK

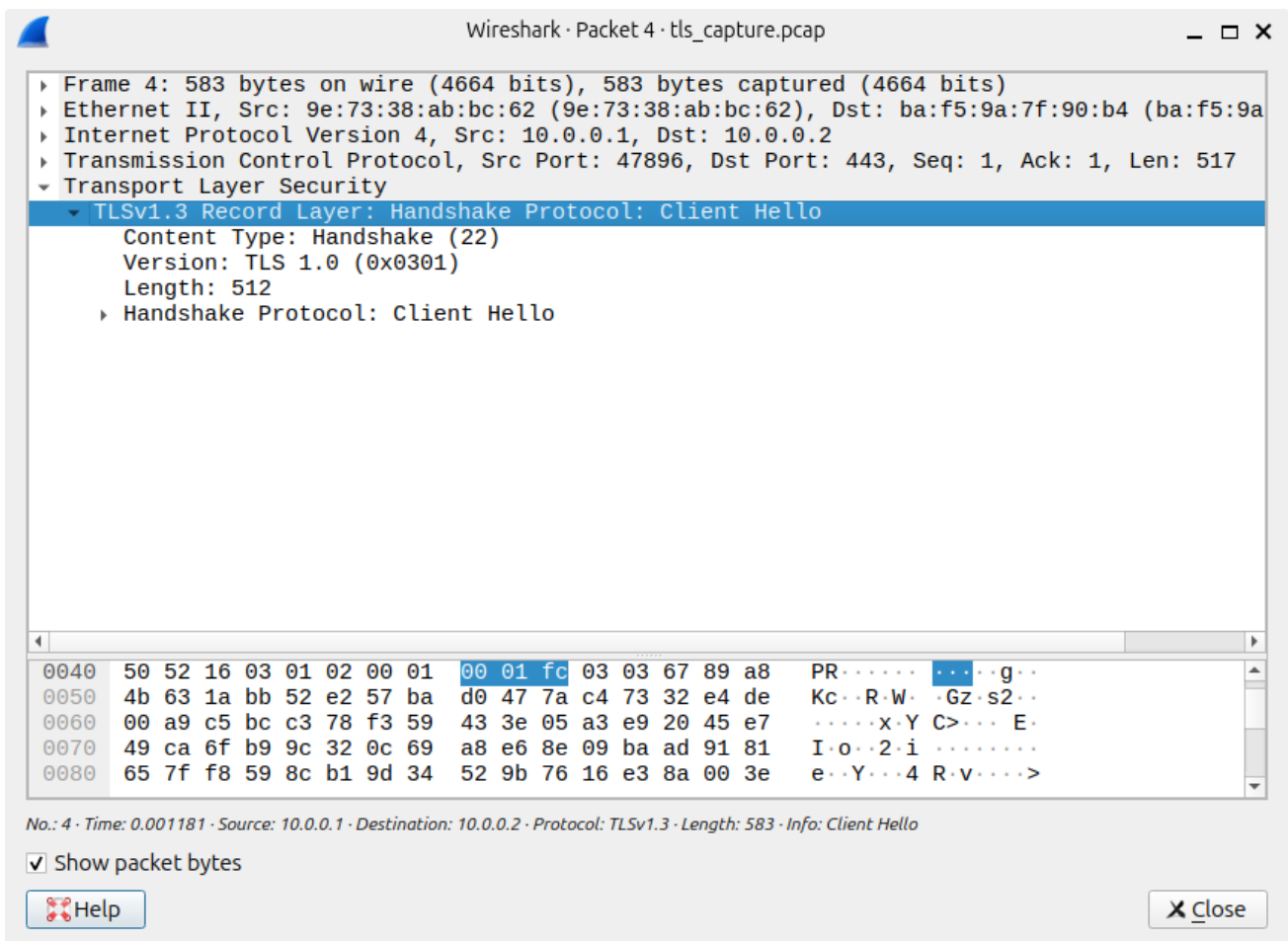


Figure 6: TLSv1.3 Client Hello

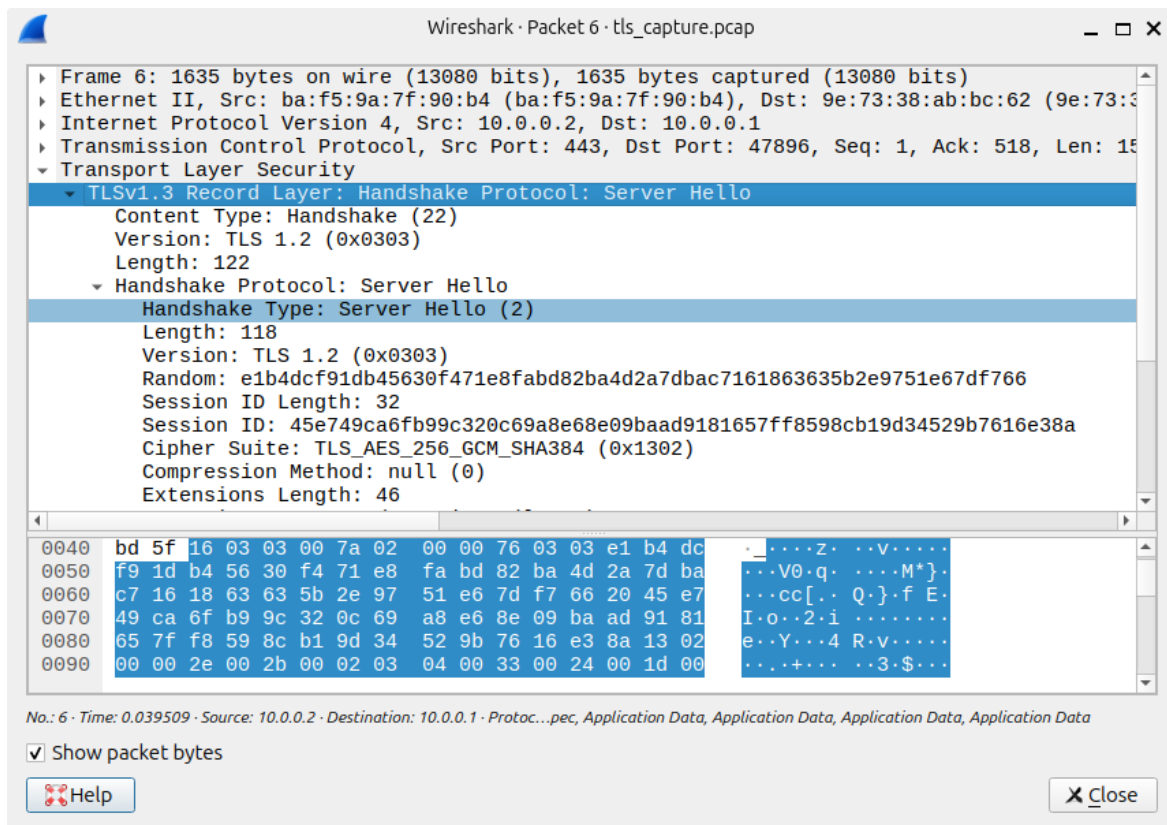


Figure 7: TLSv1.3 Server Hello

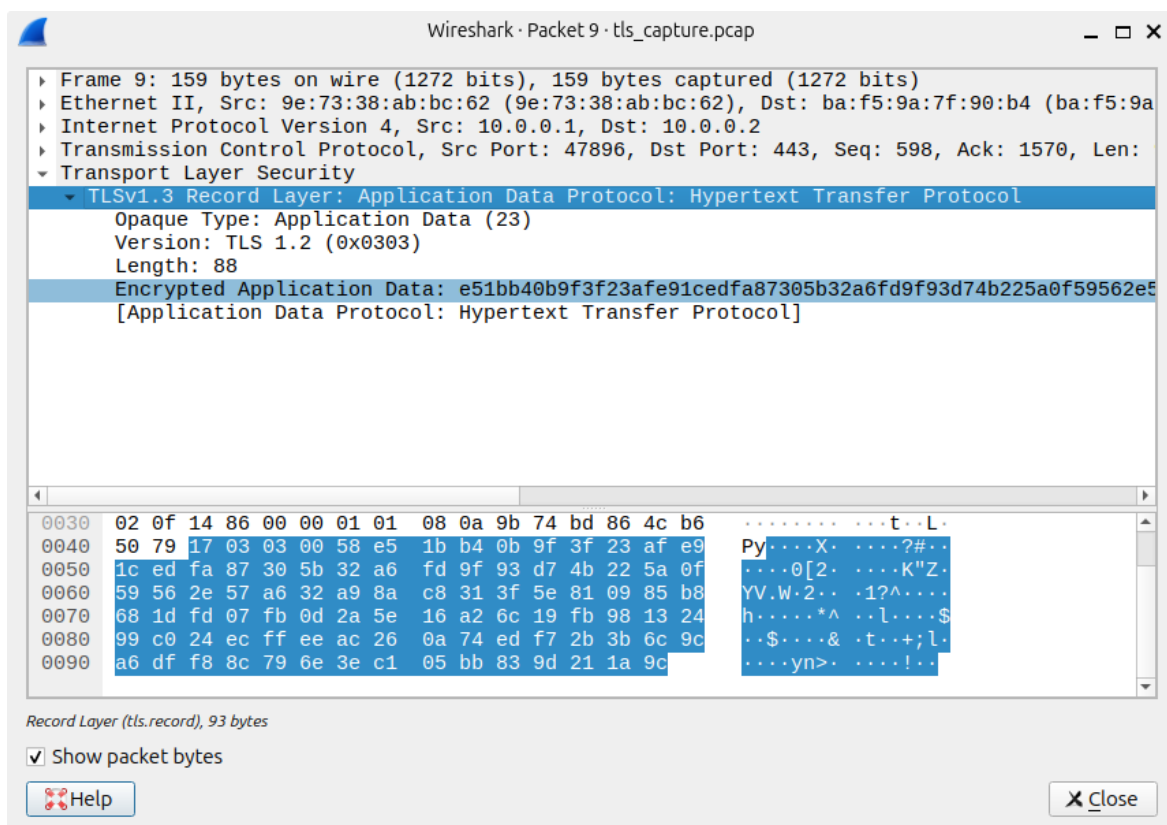


Figure 8: Encrypted Data Transfer

After Server Hello, TLS 1.3 encrypts everything.

It's actually encrypted handshake messages like: Encrypted Extensions, Certificate, Certificate Verify, Finished.

TLS 1.3 hides handshake contents.

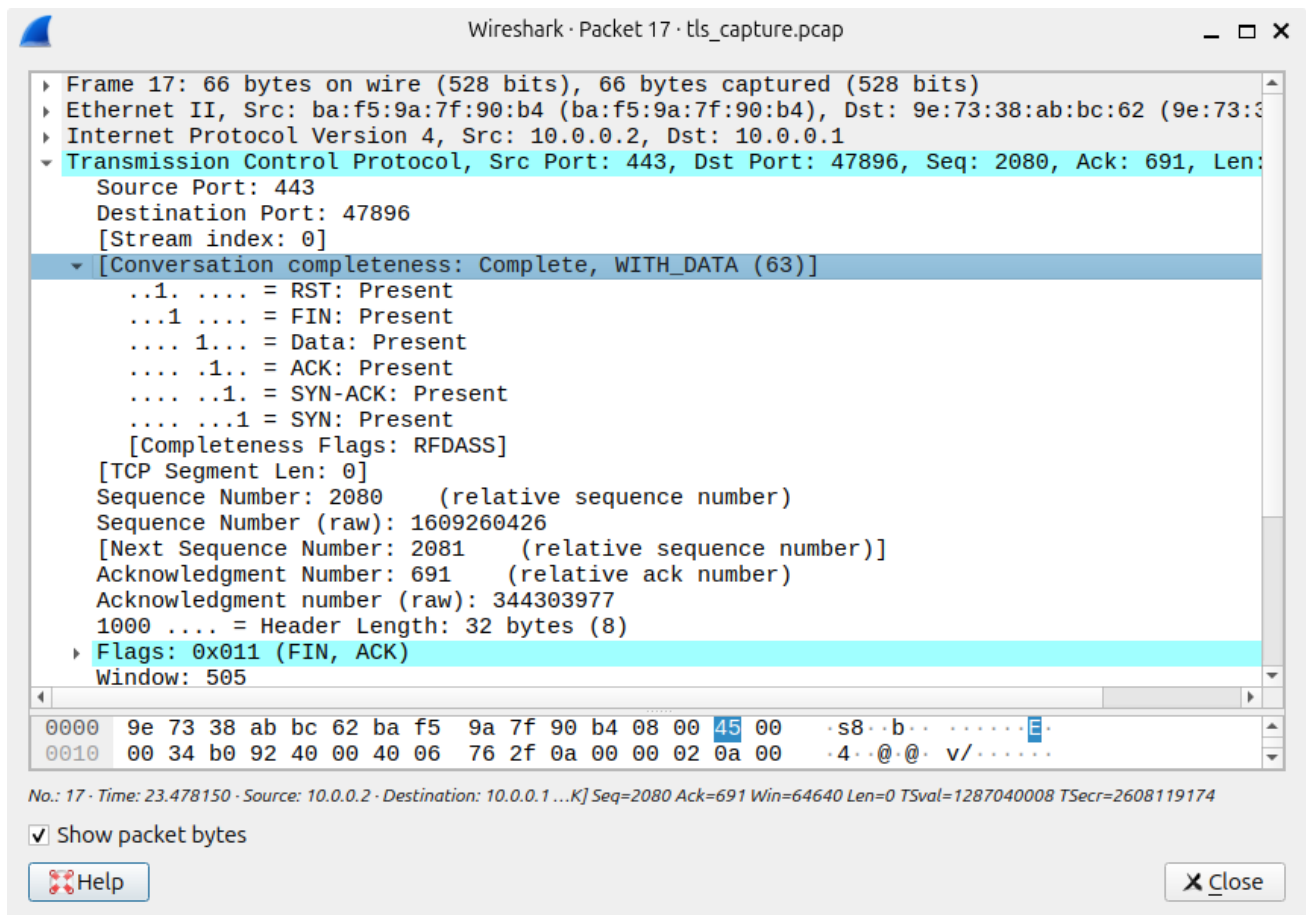
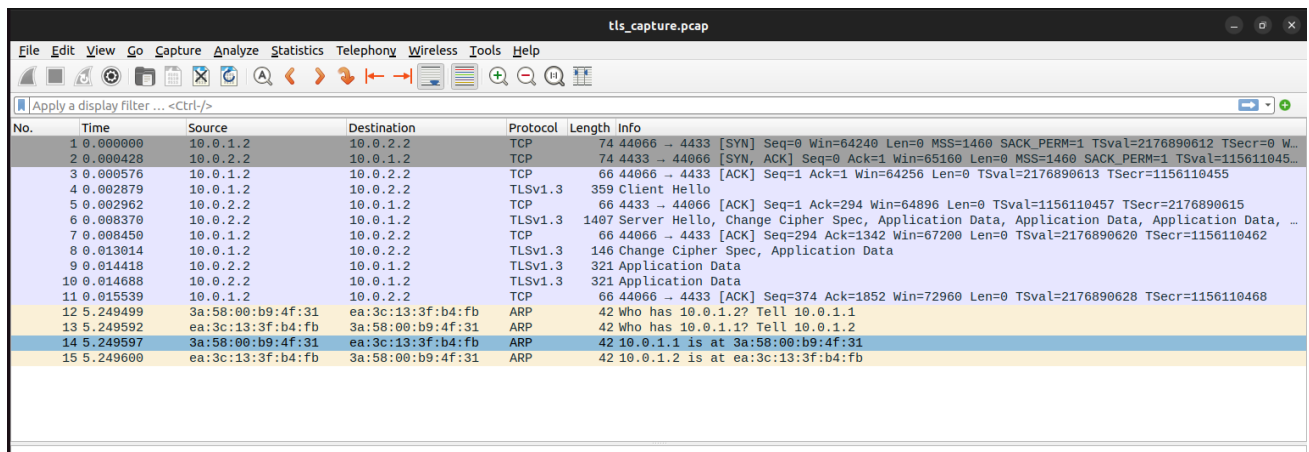


Figure 9: Connection Termination

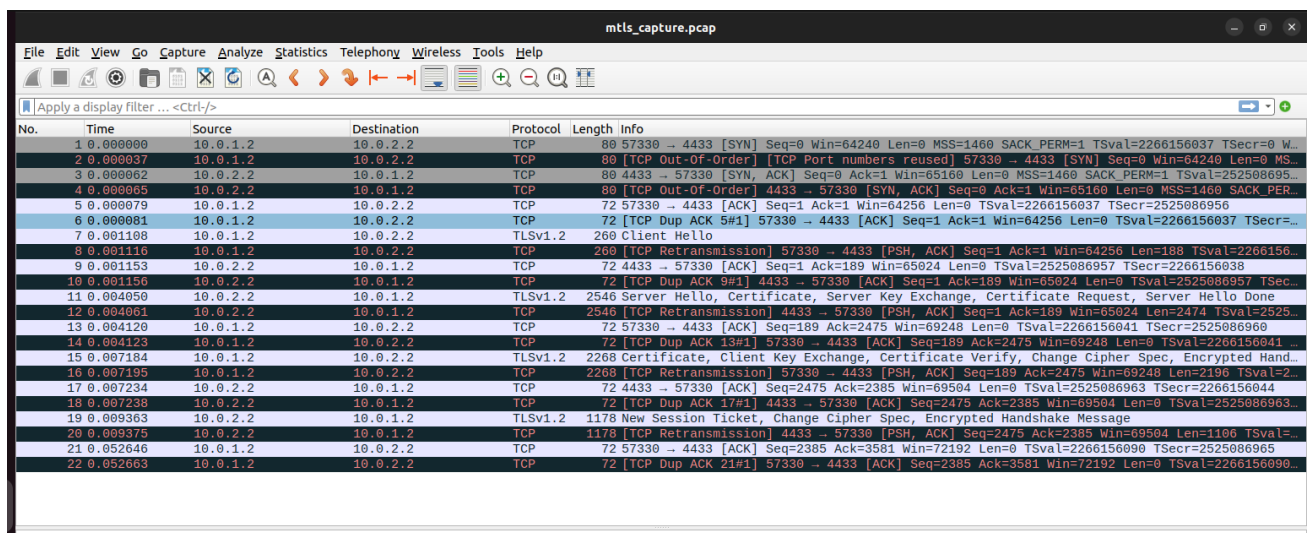
TLS vs mTLS Analysis of Wireshark



The image shows a Wireshark packet capture titled 'tls_capture.pcap'. The packet list table contains the following data:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.1.2	10.0.2.2	TCP	74	44066 → 4433 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2176890612 TSecr=0 W...
2	0.000428	10.0.2.2	10.0.1.2	TCP	74	4433 → 44066 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=2176890612 TSecr=0 W...
3	0.000576	10.0.1.2	10.0.2.2	TCP	66	44066 → 4433 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2176890613 TSecr=1156110455
4	0.002879	10.0.1.2	10.0.2.2	TLSv1.3	359	Client Hello
5	0.002962	10.0.2.2	10.0.1.2	TCP	66	4433 → 44066 [ACK] Seq=1 Ack=294 Win=64896 Len=0 TSval=1156110457 TSecr=2176890615
6	0.008370	10.0.2.2	10.0.1.2	TLSv1.3	1407	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, ...
7	0.008450	10.0.1.2	10.0.2.2	TCP	66	44066 → 4433 [ACK] Seq=294 Ack=1342 Win=67200 Len=0 TSval=2176890620 TSecr=1156110462
8	0.013014	10.0.1.2	10.0.2.2	TLSv1.3	146	Change Cipher Spec, Application Data
9	0.014418	10.0.2.2	10.0.1.2	TLSv1.3	321	Application Data
10	0.014688	10.0.2.2	10.0.1.2	TLSv1.3	321	Application Data
11	0.015539	10.0.1.2	10.0.2.2	TCP	66	44066 → 4433 [ACK] Seq=374 Ack=1052 Win=72960 Len=0 TSval=2176890628 TSecr=1156110468
12	5.249499	3a:58:00:b9:4f:31	ea:3c:13:3f:b4:fb	ARP	42	Who has 10.0.1.2? Tell 10.0.1.1
13	5.249592	ea:3c:13:3f:b4:fb	3a:58:00:b9:4f:31	ARP	42	Who has 10.0.1.2? Tell 10.0.1.2
14	5.249597	3a:58:00:b9:4f:31	ea:3c:13:3f:b4:fb	ARP	42	10.0.1.1 is at 3a:58:00:b9:4f:31
15	5.249600	ea:3c:13:3f:b4:fb	3a:58:00:b9:4f:31	ARP	42	10.0.1.2 is at ea:3c:13:3f:b4:fb

Figure 10: Pcap analysis of TLS v1.3



The image shows a Wireshark packet capture titled 'mtls_capture.pcap'. The packet list table contains the following data:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.1.2	10.0.2.2	TCP	80	57330 → 4433 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2266156037 TSecr=0 W...
2	0.000037	10.0.1.2	10.0.2.2	TCP	80	[TCP Out-Of-Order] [TCP Port numbers reused] 57330 → 4433 [SYN] Seq=0 Win=64240 Len=0 MS...
3	0.000062	10.0.2.2	10.0.1.2	TCP	80	4433 → 57330 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=2525086955 TSecr=...
4	0.000065	10.0.2.2	10.0.1.2	TCP	80	[TCP Out-Of-Order] 4433 → 57330 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PER...
5	0.000079	10.0.1.2	10.0.2.2	TCP	72	57330 → 4433 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2266156037 TSecr=2525086956
6	0.000081	10.0.1.2	10.0.2.2	TCP	72	[TCP Dup ACK 5#1] 57330 → 4433 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2266156037 TSecr=...
7	0.001108	10.0.1.2	10.0.2.2	TLSv1.2	260	Client Hello
8	0.001116	10.0.1.2	10.0.2.2	TCP	260	[TCP Retransmission] 57330 → 4433 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=189 TSval=2266156...
9	0.001153	10.0.2.2	10.0.1.2	TCP	72	4433 → 57330 [ACK] Seq=1 Ack=189 Win=65024 Len=0 TSval=2525086957 TSecr=2266156038
10	0.001156	10.0.2.2	10.0.1.2	TCP	72	[TCP Dup ACK 9#1] 4433 → 57330 [ACK] Seq=1 Ack=189 Win=65024 Len=0 TSval=2525086957 TSecr=...
11	0.004050	10.0.2.2	10.0.1.2	TLSv1.2	2546	Server Hello, Certificate, Server Key Exchange, Certificate Request, Server Hello Done
12	0.004061	10.0.2.2	10.0.1.2	TCP	2546	[TCP Retransmission] 4433 → 57330 [PSH, ACK] Seq=1 Ack=189 Win=65024 Len=0 TSval=2525086957 TSecr=...
13	0.004120	10.0.1.2	10.0.2.2	TCP	72	57330 → 4433 [ACK] Seq=189 Ack=2475 Win=69248 Len=0 TSval=2266156041 TSecr=2525086960
14	0.004123	10.0.1.2	10.0.2.2	TCP	72	[TCP Dup ACK 13#1] 57330 → 4433 [ACK] Seq=189 Ack=2475 Win=69248 Len=0 TSval=2266156041 TSecr=...
15	0.007184	10.0.1.2	10.0.2.2	TLSv1.2	2268	Certificate, Client Key Exchange, Certificate Verify, Change Cipher Spec, Encrypted Hand...
16	0.007195	10.0.1.2	10.0.2.2	TCP	2268	[TCP Retransmission] 57330 → 4433 [PSH, ACK] Seq=189 Ack=2475 Win=69248 Len=0 TSval=2266156041 TSecr=...
17	0.007234	10.0.2.2	10.0.1.2	TCP	72	4433 → 57330 [ACK] Seq=2475 Ack=2385 Win=69504 Len=0 TSval=2525086963 TSecr=2266156044
18	0.007238	10.0.2.2	10.0.1.2	TCP	72	[TCP Dup ACK 17#1] 4433 → 57330 [ACK] Seq=2475 Ack=2385 Win=69504 Len=0 TSval=2525086963 TSecr=...
19	0.009363	10.0.2.2	10.0.1.2	TLSv1.2	1178	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
20	0.009375	10.0.2.2	10.0.1.2	TCP	1178	[TCP Retransmission] 4433 → 57330 [PSH, ACK] Seq=2475 Ack=2385 Win=69504 Len=1106 TSval=...
21	0.052646	10.0.1.2	10.0.2.2	TCP	72	57330 → 4433 [ACK] Seq=2385 Ack=3581 Win=72192 Len=0 TSval=2266156090 TSecr=2525086965
22	0.052663	10.0.1.2	10.0.2.2	TCP	72	[TCP Dup ACK 21#1] 57330 → 4433 [ACK] Seq=2385 Ack=3581 Win=72192 Len=0 TSval=2266156090 TSecr=...

Figure 11: Pcap analysis of mTLS

Result

The TCP 3-way handshake and TLS 1.3 handshake between the red and blue namespaces were successfully captured and analyzed using Wireshark.